

Commercial Monitor: A Reproducible Multi-Source Framework for Research-to-Technology-to-Commercialization Forecasting

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Abstract. We present Commercial Monitor, a reproducible framework for studying whether scientific activity is followed by measurable technology adoption and commercialization-related outcomes. The design separates research emergence from downstream outcomes rather than treating publication growth as a proxy for commercial success. The initial taxonomy maps 32 arXiv categories to 14 research-backed monitoring domains and adds FinTech as a multi-source domain without a dedicated arXiv category. The runnable baseline combines publication signals with engineering and developer activity from public APIs. Forecasts use expanding-window temporal splits, calibration, simple baselines, coverage diagnostics, and null experiments. Network measures such as effective conductance are retained as transparent graph baselines rather than assumed to measure commercialization. LLM assistance is constrained to semantic annotation and reporting; schemas, targets, evaluation boundaries, and scientific decisions remain human-controlled. This paper specifies methodology and software architecture and does not assert empirical performance before the live workflow is executed.

1. Introduction

Technology forecasting can conflate scientific novelty, technical adoption, intellectual-property translation, and market success. These processes occur on different clocks and are measured by different observables. Commercial Monitor treats commercialization as a set of downstream outcomes. The primary forecasting object is $P(Y_k(t+h)=1 \mid X_t)$, where X_t contains only information observable by t and Y_k is an explicitly defined future outcome. This formulation makes temporal leakage a first-class concern.

2. Research questions

The system asks whether research-emergence signals precede technology/adoption signals; whether combinations of research, engineering, IP, labor, funding and company signals improve out-of-time prediction; whether graph measures add information beyond growth and persistence; and whether results survive source-coverage controls and null-model tests. It does not define a single commercial-value score or an investment-return forecast.

3. Taxonomy and data

The core taxonomy contains 32 arXiv categories mapped to 14 domains: AI/ML; applied mathematics; computer science; software engineering; security/cryptography; quantitative finance; quantum computing; quantum materials/electronics; complex systems; quantum simulation/ultracold matter; photonics/optics; computational science/engineering; emerging hardware/devices; and digital markets/mechanism design. FinTech is a fifteenth monitoring domain with no dedicated arXiv category. The initial live adapters use OpenAlex, GitHub and Stack Overflow. Downstream patent, funding, labor, company and market adapters are extension points.

4. Signals

Growth, z-scores, burst states, signal level and slope are transparent features. The baseline composite is $S_{-}(f,t) = \sum_j w_j x_{-}(f,t,j) / \sum_j w_j$, where x_{-} is normalized within field. Thus the index is a historical-position proxy, not an absolute market-size measure. Burst detection follows the two-state Poisson formulation of Kleinberg.

5. Networks

The current graph is a field-level growth-correlation graph, not a document-level concept co-occurrence graph. A transparent two-hop effective-conductance feature is $G_{ab} = \sum_v w_{av} w_{vb} / (w_{av} + w_{vb})$. Full weighted-network effective resistance is $R_{\text{eff}}(i,j) = L_{ii} + L_{jj} - 2L_{ij}$. These are network descriptors, not commercialization measures.

6. Prediction and evaluation

Features include research growth, technology growth, community activity, burst states, signal level, slope and graph features. Logistic regression is trained in an expanding window; for test year t only observations with year $< t$ enter training. Metrics are ROC-AUC when both classes occur, average precision and Brier score. Baselines include a training-set class prior and graph-only specifications. Null experiments permute labels within pre-defined temporal blocks.

7. Science-to-invention translation

The principal extension is explicit patent-paper linkage. Patent citations to scientific literature are an observable of science-to-invention knowledge flow, not proof of commercialization. Future data products should preserve legal status, family structure, citation lag and paper metadata separately and predict future linkage from pre-event information only.

8. Human-controlled agent boundary

LLMs may assist semantic normalization, entity resolution, taxonomy suggestions and report drafting. They should not define targets after observing outcomes, choose a winning model after seeing test results, delete observations or decide statistical significance. Operational rule: LLM = assistant, not scientific authority. Human-defined schemas, deterministic transformations, evaluation splits and published metrics remain authoritative.

9. Software and reproducibility

The stack is Python, pandas/numpy/scipy/scikit-learn, NetworkX, YAML, pytest and GitHub Actions. CSV is the initial portable artifact; Parquet/DuckDB can be introduced when volume warrants them. Deterministic sample mode separates software correctness from network availability. The workflow is currently manual so the maintainer can inspect taxonomy and code before a live run.

10. Limitations

Open bibliographic data have field and coverage biases. GitHub and Stack Overflow disproportionately represent software and English-speaking developer communities and are not universal measures for experimental physics or materials science. OpenAlex topic assignment is imperfect. APIs can be revised and have rate limits. Patent, funding, labor, company, procurement and revenue data have different licensing constraints and temporal delays. Association does not establish causal commercialization impact.

11. Expected empirical tests

The first empirical release should report source coverage, out-of-time performance against persistence and graph-only baselines, calibration, ablations, permutation/null results, sensitivity to taxonomy and source availability, and paper-patent lag distributions. No performance number is asserted before these experiments are executed.

12. Conclusion

Commercial Monitor provides a KISS, reproducible architecture for moving from research-trend detection toward empirically testable commercialization forecasting. Its central commitment is separation: research from adoption, proxies from outcomes, graph descriptors from commercial claims, and model fitting from future information.

References

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